

Matching

Example : 81k Ω versus 44k Ω ?

$$1\text{k}\Omega \times 81 = 81\text{k}\Omega$$

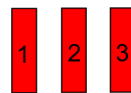
$$1\text{k}\Omega \times 44 = 44\text{k}\Omega$$

Is this the best approach?

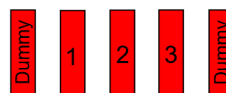
$$40\text{k}\Omega \times 2 + 1\text{k}\Omega \times 1 = 81\text{k}\Omega$$

$$40\text{k}\Omega \times 1 + 1\text{k}\Omega \times 4 = 44\text{k}\Omega$$

Moral of the story : Think Big



1, 2 and 3 are not matching



1, 2 and 3 are matching

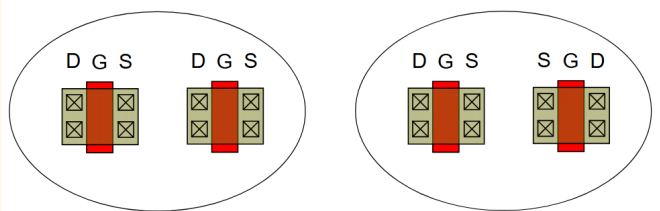
- Try to make each unit see the same surroundings

※ from manufacturing

- the implantation process might be slanted, so with shadowing effect, 1 and 2/3 is different
- then, 3 might reflect shit back to 2, so 2 and 3 is not the same

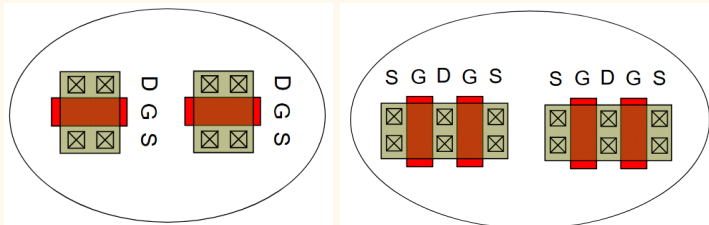
※ by using dummy, can make the manufacturing effect the same,

- the effect is by proximity, so one dummy is enough
- but if the feature size is small, need more dummies



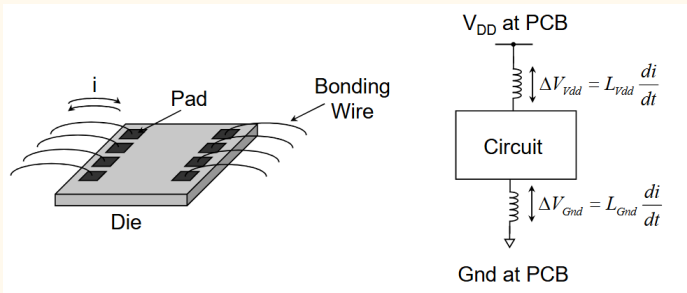
※ left is better

※ fabrication is by layer, any shift in layer will affect the left two MOS in the same way, but will make the right two MOS in a very different way (one is D bigger, the other is S bigger)

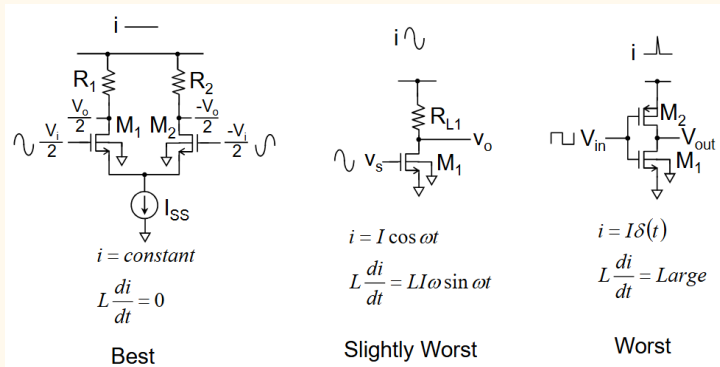


※ left is a rotated of the good layout from previous slide. with advanced process of strained technology, device orientation is important, so cannot anyhow rotate
 ※ Another way is to make the device even number fingers so the layer shift will not affect it

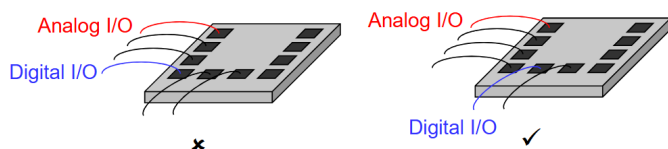
Mixed Signal



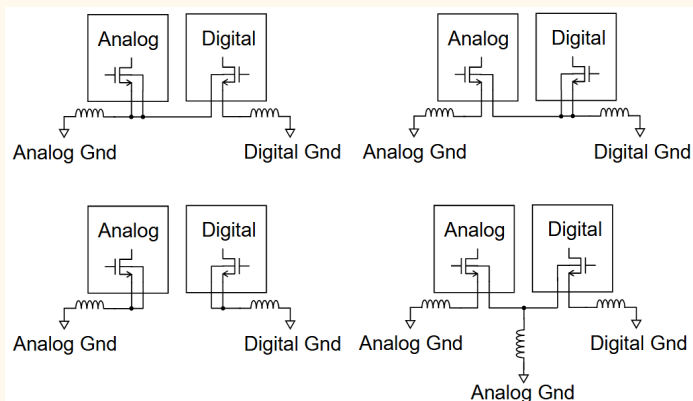
- ✂ Effect of bonding wire is the inductance, need simulate with bonding wire to make sure it will work in package
- ✂ Issue with inductance is when there is a change in supply current draw
- ✂ In QFN, there is a lot of ground bonding going to the ground pad, so a lot ground bonding without being limited by the external pins
 - $\Rightarrow$ Lower ground bound noise, but the top supply still limited so have to use more supply pad or more bonding wire per pad
 - For bonding wire in parallel, inductance in parallel decreases so $L \downarrow$. Though it does not drop by half when have 2 in parallel, due to mutual coupling



- ✂ For differential pair, the current is always constant, current increase and decrease in the two branches by the same amount
- ✂ For RF, most of the time cannot do single ended. Need to reduce parasitic or interface to antenna is single ended. But for RF the signal is small, so change in current still small
- ✂ Digital is the worst, every bit transistor is a current spike



- ✂ This is done during floor planning which is the equivalent to placement in PCB
- ✂ Noise can couple from digital to analog via capacitive or inductive coupling. To avoid inductive coupling, put on other side or orthogonal
- ✂ if no choice need put them on the same side
 - can put Vdd in between, Vdd is acting as analog ground
 - why not ground, why Vdd? In QFN, the ground is short in length, it goes straight to the bottom pad, Vdd has to go to the side pins, so can shield better
 - Digital ground should use external pad and not corrupt analog ground



- In CMOS, analog and digital circuits share the same substrate. So how to connect the substrate to not couple them?
- ✂ Out of all 4, top right is for sure wrong, the rest is up to designer
- ✂ Bottom right: for analog we want V_{BS} to be small if possible, otherwise there is body effect.
- ✂ Bottom left is relying on the finite resistance in the substrate to isolate them. Deep n-wells and guard rings can isolate the grounds.